AGENTIC-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	Amazon IAD601, VA -- station KIAD, America/New_York
Committed pair	OSM 701930920 -> 48242814 Amazon IAD98 / Amazon IAD10
Facade gap	74.1 m between the two halls
Weather record	43,763 real hourly records from KIAD
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.4086 C at 230 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2023-09-23 (safe_sector)
Plant limit	18.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 9 of 24 hours with 2 mode changes. That is 2 more than the reactive incumbent, at 0 unsafe hours against its 0. 3 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	9 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
Incumbent	7 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
	3 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	FREE	yes	-	15.857	18.0	15.560	1.152
01	FREE	yes	-	15.560	18.0	15.000	1.076
02	FREE	yes	-	15.547	18.0	14.440	1.238
03	FREE	yes	-	15.478	18.0	14.440	1.078
04	FREE	yes	-	15.270	18.0	14.440	1.064
05	FREE	yes	-	15.250	18.0	14.440	1.250
06	FREE	yes	-	14.918	18.0	13.890	0.979

07	FREE	yes	-	16.414	18.0	13.890	1.700
08	FREE	yes	-	17.782	18.0	13.890	2.078
09	mechanical	no	dry-bulb	18.841	18.0	14.440	2.391
10	mechanical	no	dry-bulb	19.067	18.0	15.000	2.244
11	mechanical	no	dry-bulb	18.234	18.0	15.000	1.783
12	mechanical	yes	switch budget	17.956	18.0	15.560	1.349
13	mechanical	yes	switch budget	17.747	18.0	15.560	1.235
14	mechanical	yes	switch budget	17.760	18.0	16.110	1.356
15	mechanical	no	dew point	17.785	18.0	16.670	1.221
16	mechanical	no	dew point	17.111	18.0	16.670	1.115
17	mechanical	no	dew point	16.827	18.0	16.110	1.385
18	mechanical	no	dew point	16.576	18.0	16.110	1.322
19	mechanical	no	dew point	16.555	18.0	16.670	1.385
20	mechanical	no	dew point	16.465	18.0	16.670	1.633
21	mechanical	no	dew point	16.291	18.0	16.110	1.708
22	mechanical	no	dew point	16.552	18.0	16.110	1.351
23	mechanical	no	dew point	16.842	18.0	16.670	1.125

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.857 C, which is 2.143 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.152 C, and the margin is measured, not chosen: 0.856 C of group-conditional forecast error for this hour of day, plus 0.2964 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

01:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.560 C, which is 2.440 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.076 C, and the margin is measured, not chosen: 0.796 C of group-conditional forecast error for this hour of day, plus 0.2794 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

02:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.547 C, which is 2.453 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.238 C, and the margin is measured, not chosen: 0.965 C of group-conditional forecast error for this hour of day, plus 0.2724 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

03:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.478 C, which is 2.522 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.078 C, and the margin is measured, not chosen: 0.880 C of group-conditional forecast error for this hour of day, plus 0.1977 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

04:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.270 C, which is 2.730 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.064 C, and the margin is measured, not chosen: 0.795 C of group-conditional forecast error for this hour of day, plus 0.2696 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

05:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.250 C, which is 2.750 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.250 C, and the margin is measured, not chosen: 0.971 C of group-conditional forecast error for this hour of day, plus 0.2794 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

06:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 14.918 C, which is 3.082 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.978 C, and the margin is measured, not chosen: 0.785 C of group-conditional forecast error for this hour of day, plus 0.1931 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

07:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.414 C, which is 1.586 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.700 C, and the margin is measured, not chosen: 1.439 C of group-conditional forecast error for this hour of day, plus 0.2614 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

08:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.783 C, which is 0.217 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.078 C, and the margin is measured, not chosen: 1.796 C of group-conditional forecast error for this hour of day, plus 0.2825 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

09:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.841 C against a 18.0 C limit, so it fails by 0.841 C. It would take a limit 0.841 C higher, or a bound 0.841 C tighter, to change this hour.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.067 C against a 18.0 C limit, so it fails by 1.067 C. It would take a limit 1.067 C higher, or a bound 1.067 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.234 C against a 18.0 C limit, so it fails by 0.234 C. It would take a limit 0.234 C higher, or a bound 0.234 C tighter, to change this hour.

12:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.956 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

13:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.747 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

14:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.760 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

15:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 17.785 C would have passed. The outside air is too HUMID: the dew-point bound is 15.28 C against a

15.0 C maximum, failing by 0.28 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

16:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 17.111 C would have passed. The outside air is too HUMID: the dew-point bound is 15.55 C against a 15.0 C maximum, failing by 0.55 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

17:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 16.827 C would have passed. The outside air is too HUMID: the dew-point bound is 15.55 C against a 15.0 C maximum, failing by 0.55 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

18:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 16.576 C would have passed. The outside air is too HUMID: the dew-point bound is 15.84 C against a 15.0 C maximum, failing by 0.84 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

19:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 16.555 C would have passed. The outside air is too HUMID: the dew-point bound is 16.11 C against a 15.0 C maximum, failing by 1.11 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

20:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 16.464 C would have passed. The outside air is too HUMID: the dew-point bound is 16.11 C against a 15.0 C maximum, failing by 1.11 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

21:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 16.291 C would have passed. The outside air is too HUMID: the dew-point bound is 16.11 C against a 15.0 C maximum, failing by 1.11 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

22:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 16.552 C would have passed. The outside air is too HUMID: the dew-point bound is 15.83 C against a 15.0 C maximum, failing by 0.83 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

23:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 16.842 C would have passed. The outside air is too HUMID: the dew-point bound is 16.12 C against a 15.0 C maximum, failing by 1.12 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

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